DATA SCIENCE
ASSIGNMENT-I

10

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Q1. Given the following data can we certain set of patients seen by a doctor. Can the doctor conclude that a person having chills, fever, mild headache and without running nose: has flu?

using naive Bayes classification,

Chills	Running nose	Headache	fever	has flu
y	N	mild	y	N
y	y	no	N	y
y	N	Strong	y	y
N	y	mild	y	y
N	N	no	N	N
N	y	Strong	y	y
N	y	Strong	N	N
y	y	mild	y	y
y	N	mild	y	?

consider the above table of patients where we can assume set of patients as $P_1, P_2, P_3 \cdots P_8$. we have to find that P_9 patient has flu or not.

we have P_9 patient has :- chills = y, fever = y, headache = mild, Running nose = N

from the informations we can calculate,

probability of patient having flu = 5/8

probability of patient not having flu = 3/8

Now we have to calculate probability based on instance

Chills	Y	N
Y	3/5	1/3
N	2/5	2/3

Runny nose	Y	N
Y	4/5	1/3
N	1/5	2/3

Fever	Y	N
Y	4/5	1/3
N	1/5	2/3

Headache	Y	N
mild	2/5	1/3
no	1/5	1/3
strong	2/5	1/3

for finding P4 has flu or not, let's calculate

$$P_{NB}(Y) = P(\text{has flu} = Y) \times P(\text{Chills} = Y) \times P(\text{Runny Nose} = N) \times P(\text{Headache} = \text{mild}) \times P(\text{Fever} = Y)$$

$$= 5/8 \times 3/5 \times 1/5 \times 2/5 \times 4/5$$

$$= 0.024$$

$$P_{NB}(N) = P(\text{Has flu} = N) \times P(\text{Chills} = Y) \times P(\text{Runny Nose} = N) \times P(\text{Headache} = \text{mild}) \times (\text{Fever} = Y)$$

$$= 3/8 \times 1/3 \times 2/3 \times 1/3 \times 1/3$$

$$= 0.004$$

Let's find that the obtained solution is correct or not

$$P(Y) = \frac{P_{NB}(Y)}{P_{NB}(Y) + P_{NB}(N)} = \frac{0.024}{0.024 + 0.004} = 0.727$$

$$P(N) = \frac{P_{NB}(N)}{P_{NB}(Y) + P_{NB}(N)} = \frac{0.004}{0.024 + 0.004} = 0.273$$

$$P(Y) + P(N) = 0.727 + 0.273 = 1 \text{ //}$$

So the obtained solution is correct. we can conclude thus,

Probability of the patient Pq is resulted as having flu with 0.727 for given instance having chills, fever, mild headache without running nose while comparing to probability of not having flu.

Q7. Consider a training data set consisting of the features of the world. Each unit has 3 features named "swim", "fly", and "crawl". Let the possible values of these features are given.

Swim - fast, slow, No

Fly - Long, short, Rarely, No

Crawl - yes, No

Sl. No	Swim	fly	crawl	class
1	Fast	No	No	Fish
2	Fast	No	yes	Animal
3	slow	No.	No	Animal
4	fast	No	No	Animal
5	No	short	No	Bird
6	No	short	No	Bird
7	No	Rarely	No	Animal
8	slow	No	yes	Animal
9	slow	No	No	Fish
10	slow	No	yes	Fish
11	No	Long	No	Bird
12	fast	No	No	Bird

each unit are classified into three - Animal, Bird and fish.

we have to find out which classification would be 13th learner under given features like swim = slow, fly = Rarely and crawl = no

probability of class = fish = 3/12

probability of class = Animal = 5/12

probability of class = Bird = 4/12

Now we have to find probability based on features

crawl	fish	Animal	Bird
yes	1/3	2/5	0/4
no	2/3	3/5	4/4

fly	fish	Animal	Bird
Long	0/3	0/5	1/4
short	0/3	0/5	2/4
no	3/3	4/5	1/4
Rarely	0/3	1/5	0/4

swim	fish	Animal	Bird
fast	1/3	2/5	1/4
slow	2/3	2/5	0/4
no	0/3	1/5	3/4

To find out which classification,

Let's find out,

$$\text{probability} - P_{nB}(\text{fish}) = P(\text{class} = \text{fish}) \times P(\text{swim} = \text{slow}) \times P(\text{fly} = \text{Rarely}) \times P(\text{crawl} = \text{no})$$

$$= 3/12 \times 2/3 \times 0 \times 2/3 = 0$$

$$P_{nB}(\text{Animal}) = P(\text{class} = \text{Animal}) \times P(\text{swim} = \text{slow}) \times P(\text{fly} = \text{Rarely}) \times P(\text{crawl} = \text{no})$$

$$= 5/12 \times 2/5 \times 1/5 \times 3/5 = 0.02$$

$$P_{nB}(Bird) = p(class = Bird) \times p(swim = slow) \times p(fly = Rarely) \times p(crawl = no)$$

$$= 4/12 \times 0 \times 0 \times 4/4 = \underline{0}$$

find the obtain of Solution is correct or not

$$P(fish) = \frac{P_{nB}(fish)}{P_{nB}(fish) + P_{nB}(Animal) + P_{nB}(Bird} = \frac{0}{0+0.02+0}$$

$$= \underline{0}$$

$$P(Animal) = \frac{P_{nB}(Animal)}{P_{nB}(fish) + P_{nB}(Animal) + P_{nB}(Bird)}$$

$$= \frac{0.02}{0+0.0.02+0} = \underline{1}$$

$$P(Bird) = \frac{P_{nB}(Bird)}{P_{nB}(fish) + P_{nB}(Animal) + P_{nB}(Bird)}$$

$$= \frac{0}{0+0.02+0} = \underline{0} \Rightarrow p(fish) + p(Animal) + p(Bird$$

$$= 0+1+0 = \underline{1}$$

So the obtained solution is correct.

probability of 13th feature is resulted as Animal classification with 1. for given features :- swim = short, fly = Rarely, crawl = No, while comparing to other classification.